

Practical 8 - Time-dependent variables

Survival Analysis

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Datasets required: `jas`, `jas1`

R packages required: `survival`, `tidyverse`

```
library(survival)
library(tidyverse)
```

Introduction

In this practical we will perform analyses of data from the Stanford Heart Transplant programme. We saw examples from this in the lecture. The data records information on individuals in a programme in which they were waiting for a heart transplant. It includes 103 individuals who were followed up from the date of their acceptance onto the programme to death or censoring. A version of this classic data set is available in the `survival` package in R. We will use the versions called `jas` and `jas1`, where the name refers to the *Journal of the American Statistical Association* where results from analysis of the data were published (e.g. Crowley and Hu (1977)).

Variables we will use in the `jas` data (one row per individual) are:

Variable	Description
<code>futime</code>	Time in days to death or censoring.
<code>fustat</code>	Status at the end of follow-up. Death: 1, Censoring: 0
<code>transplant</code>	Indicator of whether the person ever received a transplant
<code>wait.time</code>	Time in days to transplant
<code>age</code>	Age in years at joining the programme
<code>surgery</code>	Prior bypass surgery (No: 0, Yes: 1)
<code>tx.date</code>	Transplant date

Variables we will use in the `jasal` data (more than one row per individual) are:

Variable	Description
<code>id</code>	Patient id
<code>start</code>	Start time for this time interval
<code>stop</code>	Stop time for this time interval
<code>event</code>	Status at the end of the interval. Death: 1, Censoring: 0

Aims

The aims of this session are:

- Use data sets with one row per individual and multiple rows per individual for analysis involving time-dependent variables.
- Understand how immortal time bias can arise.
- Fit and interpret Cox models including time-dependent explanatory variables.
- Use the landmarking approach for analysis of time-dependent variables.

Question 1

We will begin by investigating what is in the data and how the data are arranged.

- Load the datasets `jasal` and `jasal`. How many records (rows) are there in the two datasets? And how many individuals?
- How many individuals received a transplant?
- Look at the data for individuals 6-10 in the two data sets and see how the variables `wait.time`, `futime`, `transplant` in `jasal` are connected to the variables `start`, `stop` and `transplant` in `jasal`.

Question 2

- Using the data with one row per individual (`jasal`) perform a Kaplan-Meier analysis by transplant status, and perform a Cox regression to estimate the hazard ratio for transplant.
- What is wrong with these analyses? How long are individuals 7 and 10 ‘immortal’ in the way that the data are being used here?

Question 3

- We will now consider transplant as the time-dependent variable that it is. Write down the form of a Cox proportional hazards model for death, with transplant status as a time-dependent variable.
- Fit the model using the data in the long format (`jasal`) and interpret the results. Think about what the format of the `Surv()` function should be to account for the long format of the data.

Question 4

In this question we will extend the model in question 3 to allow the hazard to depend not only on current transplant status, but also on the time since transplant. Use the following steps:

- Start by splitting the long form data (`jasal`) using the unique event times as the cut points. You can do this using `survsplit`, which we have used in earlier practicals.
- Generate a time-dependent variable that represents the time since transplant in each row, setting this to 0 prior to transplant.
- Categorise the time since transplant into three periods: period 1: 0-90 days, period 2: 91-186 days, period 3: >186 days
- Fit a Cox model using time-dependent transplant status plus its interaction with the time period.
- Discuss how the association between transplant and the hazard depends on time-since-transplant.

Question 5

In this question we will apply the landmarking approach, using a landmark time of 90 days. Go through the following steps:

- Using the `jasal` dataset, restrict the data to people who remain alive 90 days after joining the programme.
- Reset the time origin to be day 90. This means that you need to modify the follow-up time variable `futime` to reflect the change in the time origin.
- Generate a variable that indicates whether or not a person received a transplant before day 90, i.e. their transplant status at the landmark time.
- Generate Kaplan-Meier survival curves by transplant status at the landmark time based on this modified data, and fit a Cox model.
- What happens if you change the landmark time to 30 days?

Question 6

Suppose we are interested in studying post-transplant survival, and how the following factors are associated with post-transplant survival: age (**age**), whether or not the person had had prior bypass surgery (**surgery**), calendar year in which the transplant was received (using **tx.date**).

- Starting from the **jasa** data, set-up a new dataset that enables you to study post-transplant survival.
- Obtain a Kaplan-Meier plot for post-transplant survival. What are the estimated probabilities of survival to 30 days and 1 year post-transplant?
- Fit a Cox model including **age**, **surgery** and calendar year of transplant. Interpret the results.
- Investigate the assumptions of the model using Schoenfeld residuals and Martingale residuals.